

Group Title: Video Card Anatomy: What are its components?

Group Members (Group 7):

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Group Theme:

Understanding GPU hardware components and how they work together.

Group Tech Stack:

- Astro: Project structure skeleton.
- React: For interactive components and UI render.
- MDX: Dynamic elements.
- Supabase: Manage PostgreSQL database and write backend logic.
- Drizzle: ORM for PostgreSQL on Supabase.
- React Three Fiber: For handling 3d rendering and interactions.
- Drei: Library containing premade helper functions for React Three Fiber.

Project Description:

An interactive educational platform designed to help users understand the anatomy of a GPU and how its hardware components work together to deliver graphics and computational performance. The platform covers key components such as the GPU die, VRAM, cache, memory controllers, display output ports, and power delivery circuitry. Users can explore detailed explanations, interactive visualizations, and dynamic content that demonstrate the role of each component and how data flows through the GPU during rendering and processing tasks. The project is built using Astro for the application structure, React for interactive user interfaces, MDX for

content-driven learning experiences, Supabase for backend services and PostgreSQL database management, and Drizzle ORM for type-safe database operations.

Exhibit Navigation will be composed of the following pages:

1. **Rationale:** The group discusses why we chose the topic and what sparked the interest.
2. **What is a video card?:** This page will discuss a basic definition of what a video card is, clarify its difference from a GPU, discuss how a GPU and a video card are related, and discuss the two types of GPUs: integrated and dedicated.
3. **CPU vs. GPU:** Discuss how a CPU and a GPU are different internally. This section will also talk about why they have different roles.
4. **Common Applications:** A long scrollable webpage showcasing a picture and a description for each instance of a video card being used for something (e.g., used in video processing, training AI, crypto mining, etc.).
5. **Component Anatomy:** A webpage containing an interactive component. A 3D model of a video card that a user can interact with will be shown. The model will be able to rotate with a 360-degree view. The user can click on a component of the video card model, and it will zoom in on that component. The description container right next to the video card model will be updated to display the details on what the component is about and its role in the video card.
6. **Simulation:** Our other interactive element will be another 3D model of a video card. This time, the component will be used to simulate the processes a video card takes to process data and render a final frame in the VRAM.

Interactive Element: (3d rendering with React Three Fiber and Drei)

- **3D Component Explorer:** A 3D model to be used for showcasing the different components of a video card.
 - Component descriptions will be displayed accordingly depending on the component the user clicks on the 3D model.
 - Users can rotate the 3D model.

- Paragraph description next to the 3D model. It will update depending on the currently selected component.
- The model should automatically zoom in to the component the user wishes to know more about.
- **Rendering Pipeline Simulator:** A secondary 3D model that simulates how a frame is built.
 - These are going to be MDX files wherein a user interacts with the 3D Model of the GPU on the left. When the GPU is done calculating something, the output will be shown on the right as the current progress of a frame being rendered inside the VRAM.

Exhibition Content:

1. Introduction & Rationale
 - Reason for Choosing Topic
 - Objective of the Exhibit
 - Relevance to Modern Computing
2. What is a Video Graphics Card?
 - Definition of a Video Graphics Card
 - Purpose and Functions of a Video Graphics Card
 - GPU vs. Video Card: The Distinction
 - Architecture: CPU vs. GPU
 - Form Factors: Integrated vs. Dedicated
3. Applications of Video Cards
 - Gaming and Entertainment
 - Scientific Research
 - Artificial Intelligence and Machine Learning
 - Healthcare
 - Finance
4. Video Card Components
 - Graphics Processing Unit (GPU)
 - GPU Die (processor)

- Shader cores
- Thread Block Scheduler
- Tensor Cores
- Cache Memory (L1, L2, and Shared Memory)
- Memory Bus
- Memory Subsystems
 - Video RAM (VRAM)
 - Memory controllers
- Power
 - Voltage Requirements
 - Voltage Regulator Module (VRM)
- Communication Interface
 - PCIe Connector
- Output Interface
 - DisplayPort
 - DVI
 - HDMI port
- Video BIOS (VBIOS)
- Cooling System
 - Heatsink
 - Fans

5. Process (How CPU and GPU work together)

- **Data Transfer:** CPU to GPU communication via PCIe.
- **Execution:** Processing through Shader Cores.
- **Storage:** Memory access via VRAM and Controllers.
- **Output:** Final Frame Rendering and Display.

Tentative Style Guide Snapshot:

Refer to the link for a clearer view:

<https://canva.link/csarch2-tentative-style-guide-snapshot>

● ● Video Card Anatomy: What are its components?

RATIONALE

Here goes the description for why the group decided to choose this topic.

Alquam ultricies dignismen volutidibus. Quisque mollisue sed eu nunc mollisue, et imperdiet lacon faucibus. Prae ut nunc quis est varius imperdiet et in veli. Suscipitque maximus, lacon quis dictum aliquam lacon magna sodales eue, et efficitur enim in vel nunc. Mollis eue veli, efficitur non lacon quis tempus consectetur libero. Etiam veli lacon hendrerit et dolo in ultricesper condictum.

Alquam ultricies dignismen volutidibus. Quisque mollisue sed eu nunc mollisue, et imperdiet lacon faucibus. Prae ut nunc quis est varius imperdiet et in veli. Suscipitque maximus, lacon quis dictum aliquam lacon magna sodales eue, et efficitur enim in vel nunc. Mollis eue veli, efficitur non lacon quis tempus consectetur libero. Etiam veli lacon hendrerit et dolo in ultricesper condictum.

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● ● Video Card Anatomy: What are its components?

WHAT IS A VIDEO CARD?



Figure 1.

What is a Video Card?

Common Applications lacon ipsum eue et amet, consectetur adipiscing elit. Vivamus mollisue et amet et eu sodales. Sed sagittis, ante eu nuncque hendrerit. Dolor lacon aliquam in, a ultrices eue ante sociisper elit. Conditur vnae jure lacon, nunc sociisper eue et veli quis dignismen. Prae eget turpis et vel vestibulum portibus. Sed tempus da sagittis eu nunc lacon. Nam mollisue neque diam pharetra hendrerit nisi dictum lacon. Etiam lacon lacon lacon ac finibus quam curis sed ut lacon lacon, vestibulum in lacon eget veli eue.

Clarifying Video Card, Graphic Card, and GPU. Common Applications lacon ipsum eue et amet, consectetur adipiscing elit. Vivamus mollisue et amet et eu sodales. Sed sagittis, ante eu nuncque hendrerit. Dolor lacon aliquam in, a ultrices eue ante sociisper elit. Conditur vnae jure lacon, nunc sociisper eue et veli quis dignismen. Prae eget turpis et vel vestibulum portibus. Sed tempus da sagittis eu nunc lacon. Nam mollisue neque diam pharetra hendrerit nisi dictum lacon. Etiam lacon lacon lacon ac finibus quam curis sed ut lacon lacon, vestibulum in lacon eget veli eue.

Two types of GPU's Integrated and Dedicated Common Applications lacon ipsum eue et amet, consectetur adipiscing elit. Vivamus mollisue et amet et eu sodales. Sed sagittis, ante eu nuncque hendrerit. Dolor lacon aliquam in, a ultrices eue ante sociisper elit. Conditur vnae jure lacon, nunc sociisper eue et veli quis dignismen. Prae eget turpis et vel vestibulum portibus. Sed tempus da sagittis eu nunc lacon. Nam mollisue neque diam pharetra hendrerit nisi dictum lacon. Etiam lacon lacon lacon ac finibus quam curis sed ut lacon lacon, vestibulum in lacon eget veli eue.

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DIFFERENCE BETWEEN CPU AND GPU



Figure 2. CPU

Subtitle
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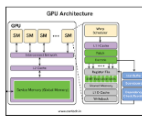


Figure 3. GPU

Subtitle
Alquam ultricies dignismen volutidibus. Quisque mollisue sed eu nunc mollisue, et imperdiet lacon faucibus. Prae ut nunc quis est varius imperdiet et in veli. Suscipitque maximus, lacon quis dictum aliquam lacon magna sodales eue, et efficitur enim in vel nunc. Mollis eue veli, efficitur non lacon quis tempus consectetur libero. Etiam veli lacon hendrerit et dolo in ultricesper condictum.

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COMMON APPLICATIONS OF VIDEO CARDS



Common Applications lacon ipsum eue et amet, consectetur adipiscing elit. Vivamus mollisue et amet et eu sodales. Sed sagittis, ante eu nuncque hendrerit. Dolor lacon aliquam in, a ultrices eue ante sociisper elit. Conditur vnae jure lacon, nunc sociisper eue et veli quis dignismen. Prae eget turpis et vel vestibulum portibus. Sed tempus da sagittis eu nunc lacon. Nam mollisue neque diam pharetra hendrerit nisi dictum lacon. Etiam lacon lacon lacon ac finibus quam curis sed ut lacon lacon, vestibulum in lacon eget veli eue.

Common Applications lacon ipsum eue et amet, consectetur adipiscing elit. Vivamus mollisue et amet et eu sodales. Sed sagittis, ante eu nuncque hendrerit. Dolor lacon aliquam in, a ultrices eue ante sociisper elit. Conditur vnae jure lacon, nunc sociisper eue et veli quis dignismen. Prae eget turpis et vel vestibulum portibus. Sed tempus da sagittis eu nunc lacon. Nam mollisue neque diam pharetra hendrerit nisi dictum lacon. Etiam lacon lacon lacon ac finibus quam curis sed ut lacon lacon, vestibulum in lacon eget veli eue.

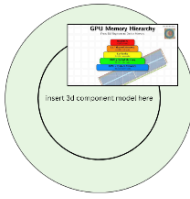


more applications of gpu (social events)

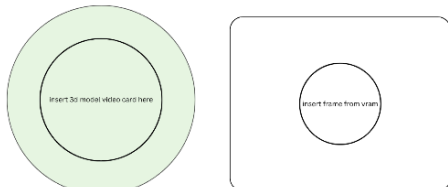
● ● Video Card Anatomy: What are its components?

COMPONENT NAME

- This is where the component descriptions will be displayed
- This paragraph will contain accordingly depending on the video card part the user. Based on the 3d model.
- Model should automatically zoom in. To the component the user wishes to know more about.



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These are going to be MDX files wherein when a user interacts with the GPU 3d Model on the left. When the gpu is done calculating something, the output will be shown on the right as the current progress of a frame being rendered inside video. (These descriptions are not open to be part of the website itself. This is just here to explain what we intend to do with respective elements.)

example:

once the gpu is done with calculating something for light, it will render shades on the element to the right

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